

A Packing of 10 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 10 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.076624. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.045871, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 10 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.076624 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 10$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 3.045871, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 10/9 = 3.076923$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N10.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
307662410915314777401039774804272227998651009829195533474766681049149644
540368785548872867727587831239952417825511348618844439601918554670659141
2269053536872519421501757093089716916117008201337
```

denominator:

```
999999999998523952536970369417107508871538705562184773729736918874371349
893162136965685789880482499246513507300509970070378479406069516021848798
468770357989490306694447047257131199721646518114
```

B Coordinates

10 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N10.txt`.

```
-315203692588 38512738141 -1736989035 136155924779 20232249973 -366955245177 285402037466 401360418627 27696206633 78799281779 -427143671995 189330207770
-14050855963 389401538640 677144636469 162166744939 32016590959 -150686983379 312490837824 -435444463965 1597243006 -15679706139 284128786863 -53087767247
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